

UMC 110nm Logic and Mixed-Mode AI Advanced Enhancement 2.5V, Operated at 3.3V, PCAP SPICE Model Document

**Version 0.2 Phase 1
2012/05/15**

Table of Contents

1	INTRODUCTION	5
1.1	Revision History	5
1.2	General Information	6
1.3	Model Usage.....	6
2	CAPACITANCE MODEL.....	9
2.1	Silicon Wafer for Modeling.....	9
2.1.1	Test devices	10
2.1.2	Measurement conditions.....	11
2.2	Electrical Parameters	12
2.2.1	Cg vs. Vg plots	13
2.2.2	Corner Cg vs. Vg plots	15
2.3	Equation to Estimate Maximum Capacitance	17
3	APPENDIX.....	18
3.1	HSPICE Warning Message	18
3.2	SPECTRE Warning Message	18
3.3	ELDO Warning Message	18

List of Figures

Figure 1-1 PCAP layout	8
Figure 2-1 Cg vs. Vg at 25 °C	14
Figure 2-2 Corner Cg vs. Vg	16

List of Tables

Table 1-1 Revision history	5
Table 2-1 Test wafer information for PCAP	9
Table 2-2 Geometry of PCAP devices measured for parameters extraction.....	10
Table 2-3 Test conditions for Cg curves of PCAP	11
Table 2-4 Electrical Parameters.....	12

1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2012/05/15	Chen Hua Tsai	Origin release

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

UMC 0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor AI
Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.9
Cadence SPECTRE version 10.1.1.111.isr8 32bit
Mentor Graphics ELDO version v6.11_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

TT: Typical PCAP case
FF: Maximum PCAP case
SS: Minimum PCAP case

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator.
.lib "/ModelFileName.lib" CaseName
, where CaseName could be TT, FF, or SS.
- Define device condition by 'Xxx' statement.

Example:

X1 VGG GND DeviceName LF=2E-6 WF=5E-6 NF=2

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation

directory.

- Add the following statement in the input file to the simulator.

include "./ModelFileName.lib.scs" section=CaseName

, where CaseName could be TT, FF, or SS.

- Define device condition by 'Xxx' statement.

Example:

X1 (VGG GND) DeviceName LF=2E-6 WF=5E-6 NF=2

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.

- Add the following statement in the input file to the simulator.

.lib "./ModelFileName.lib.eldo" CaseName

, where CaseName could be TT, FF, or SS.

- Define device condition by 'Xxx' statement.

Example:

X1 VGG GND DeviceName LF=2E-6 WF=5E-6 NF=2

ModelFileName: l110ae_pcap25od33_v021

DeviceName: pcap_25od33

Geometry description (Before shrink):

$0.25\mu\text{m} \leq \text{LF} \leq 2\mu\text{m}$

$2.5\mu\text{m} \leq \text{WF} \leq 10\mu\text{m}$

$2 \leq \text{NF} \leq 8$

Voltage Range:

$-3.3\text{V} (*1.1) \leq \text{VGS} \leq 3.3\text{V} (*1.1)$

Temperature range:

$-50^{\circ}\text{C} \sim +125^{\circ}\text{C}$

Layout example:

The following figure is the layout example of the PCAP. LF and WF are the single poly finger length and width, respectively, and Nf is the total finger number.

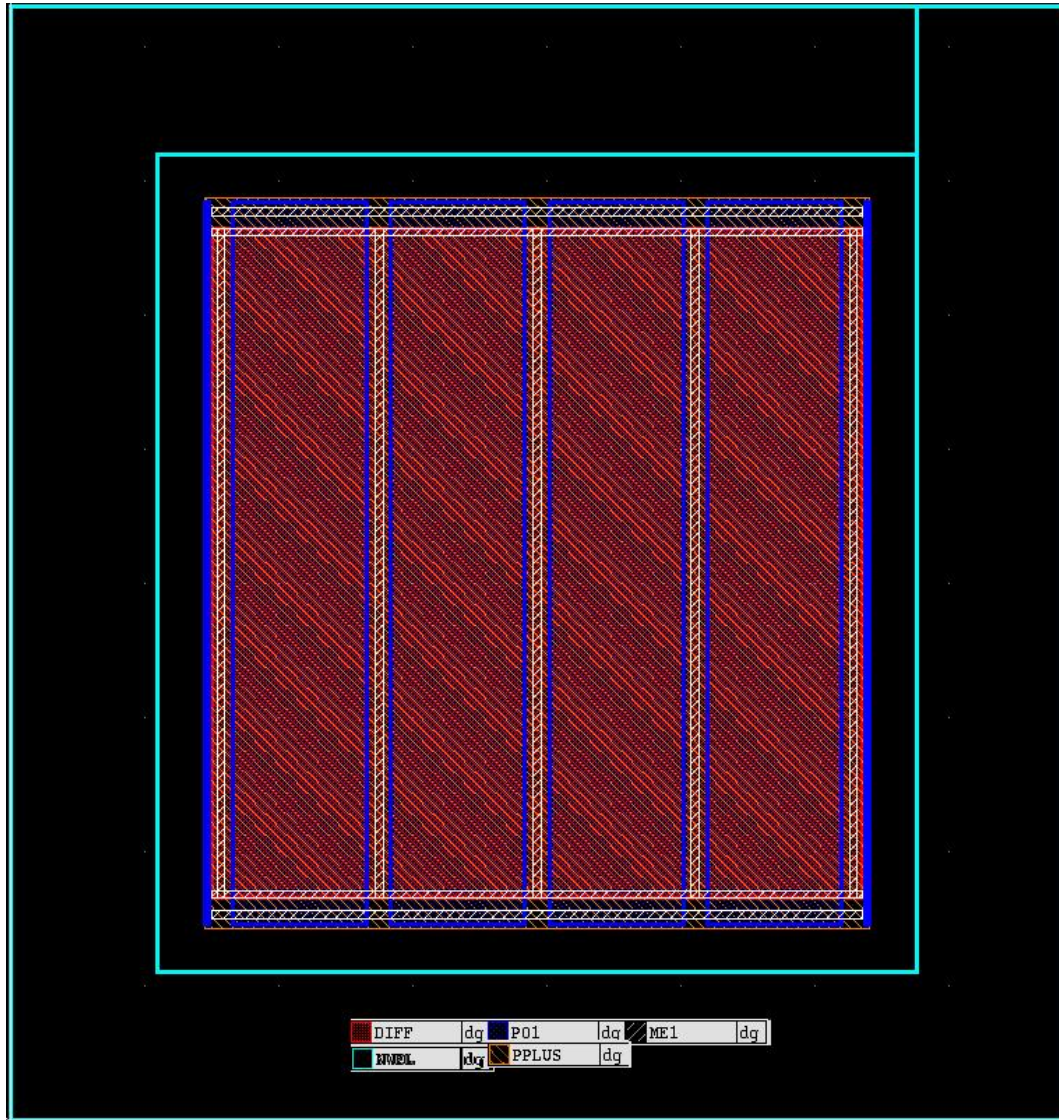


Figure 1-1 PCAP layout

2 Capacitance Model

2.1 Silicon Wafer for Modeling

The information of the hardware used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for PCAP

	Wafer for PCAPs measurements data
Mask ID	A11007
Lot ID	D5FHR
Wafer Number	#11

2.1.1 Test devices

The following geometries were available for parameters extraction.

Table 2-2 Geometry of PCAP devices measured for parameters extraction.

W\Lf	2	1	0.5	0.35	0.25
10			X		
5	X	X	X	X	X
2.5			X		

2.1.2 Measurement conditions

The capacitance was measured under the conditions summarized in the table below.

Table 2-3 Test conditions for Cg curves of PCAP

$V_{DD} = -3.3V$	C_G vs. V_G
V_G	From -3.63 to 3.63 V in 0.066 V step
V_B	0V
Temp [°C]	-50,25,125

2.2 Electrical Parameters

The following table shows the measured and simulated Cmax (fF/um²), where Cmax is the capacitance at VG=-3.3V, VB=0 V, T=25°C.

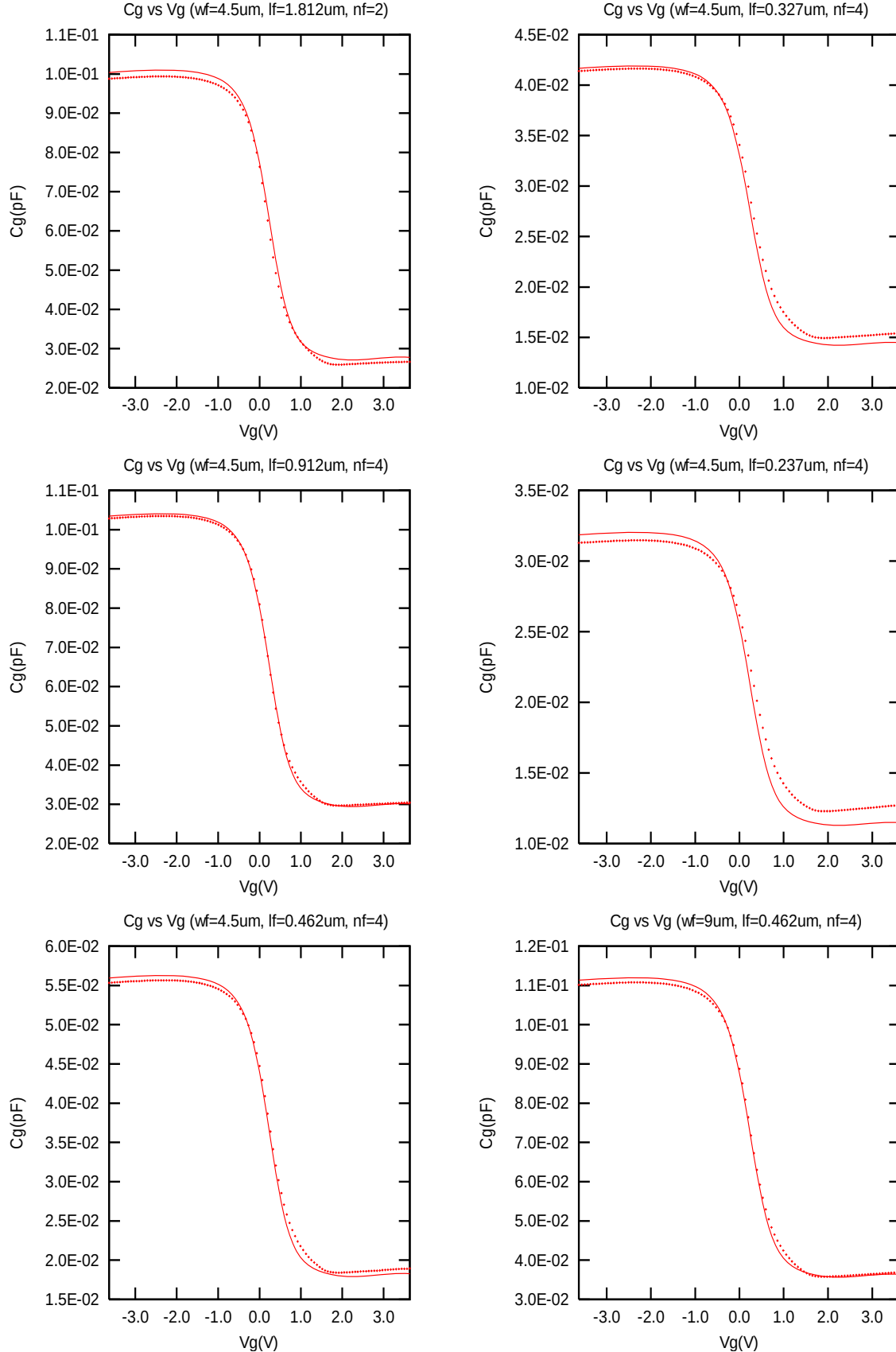
Table 2-4 Electrical Parameters

Wf*0.9 (um)	Lf*0.9 +0.012 (um)	nf	Target (as-centered) (fF/um ²)	Measured Cmax* (fF/um ²)	Model Cmax* (as-extracted) (fF/um ²)	Model Cmax* (as-centered) (fF/um ²)
4.5	1.812	2	6.395	6.070	6.169	6.5
4.5	0.912	4		6.280	6.316	6.656
4.5	0.462	4		6.668	6.742	7.104
4.5	0.327	4		7.047	7.095	7.476
4.5	0.237	4		7.348	7.484	7.886
9	0.462	4		6.635	6.708	7.068
4.5	0.462	8		6.702	6.742	7.104
2.25	0.462	4		6.824	6.880	7.249
4.5	0.462	2		6.696	6.742	7.104

*Cmax is normalized by the area (Wf*0.9 and Lf*0.9+0.012) after shrinkage.

2.2.1 Cg vs. Vg plots

The following figures show the measured and simulated Cg (pF) vs. Vg (V) after shrinkage. The measurement data is dots and simulation is line, respectively.



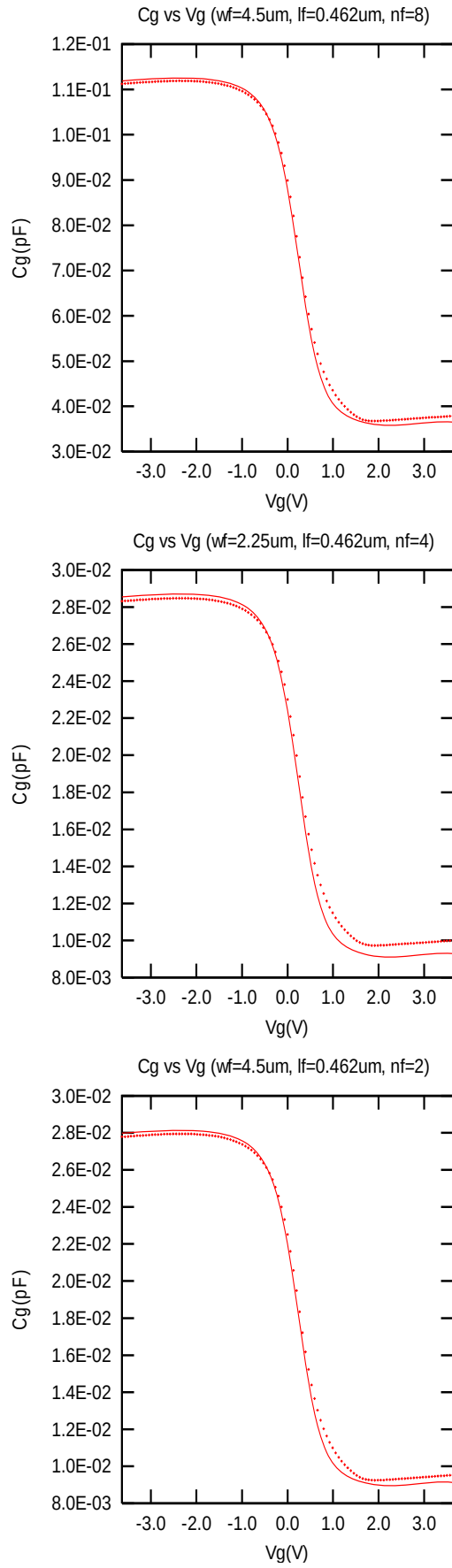
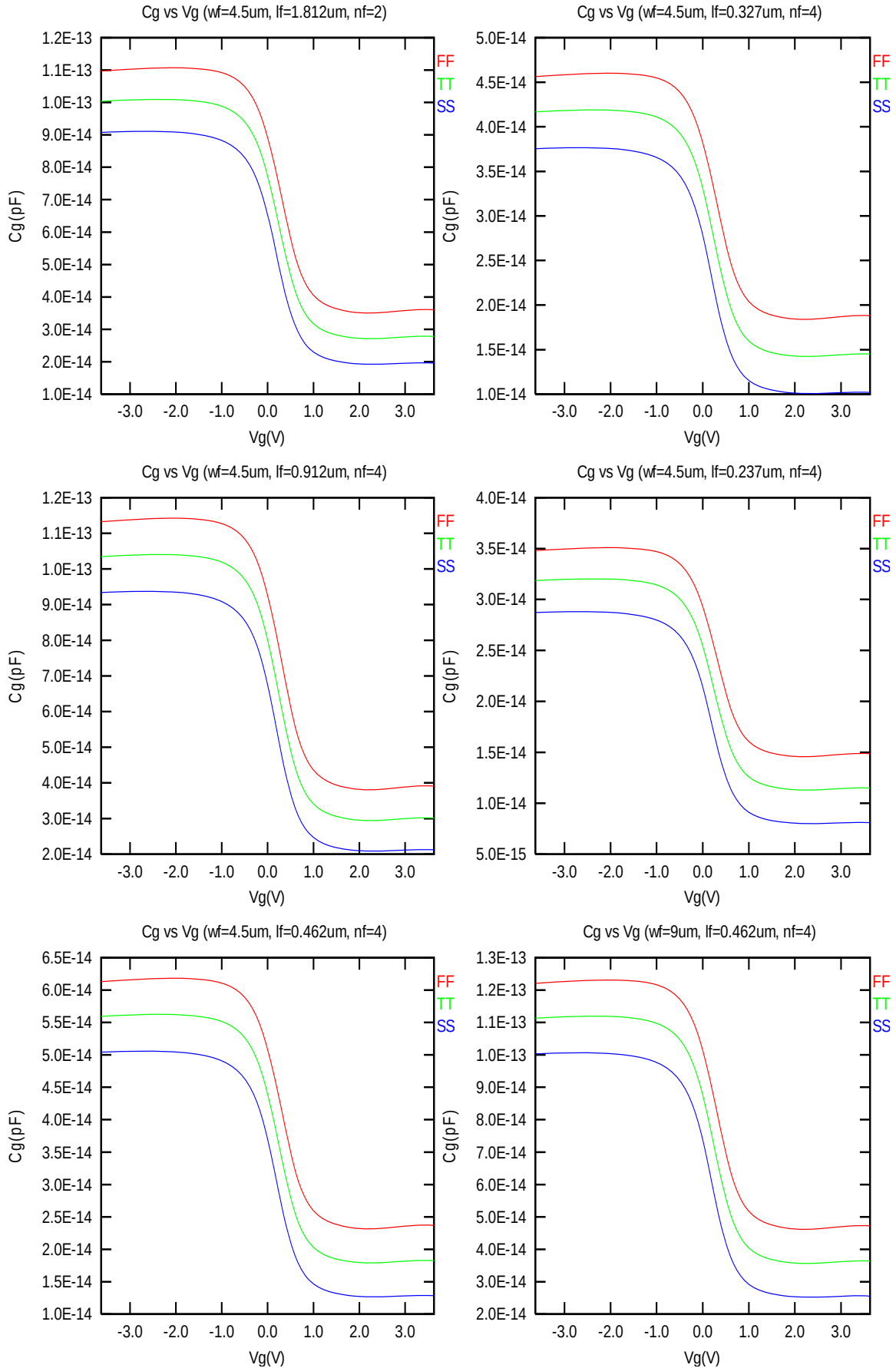


Figure 2-1 Cg vs. Vg at 25 °C

2.2.2 Corner Cg vs. Vg plots

The following figures show the simulated corner Cg (pF) vs. Vg (V) plots.



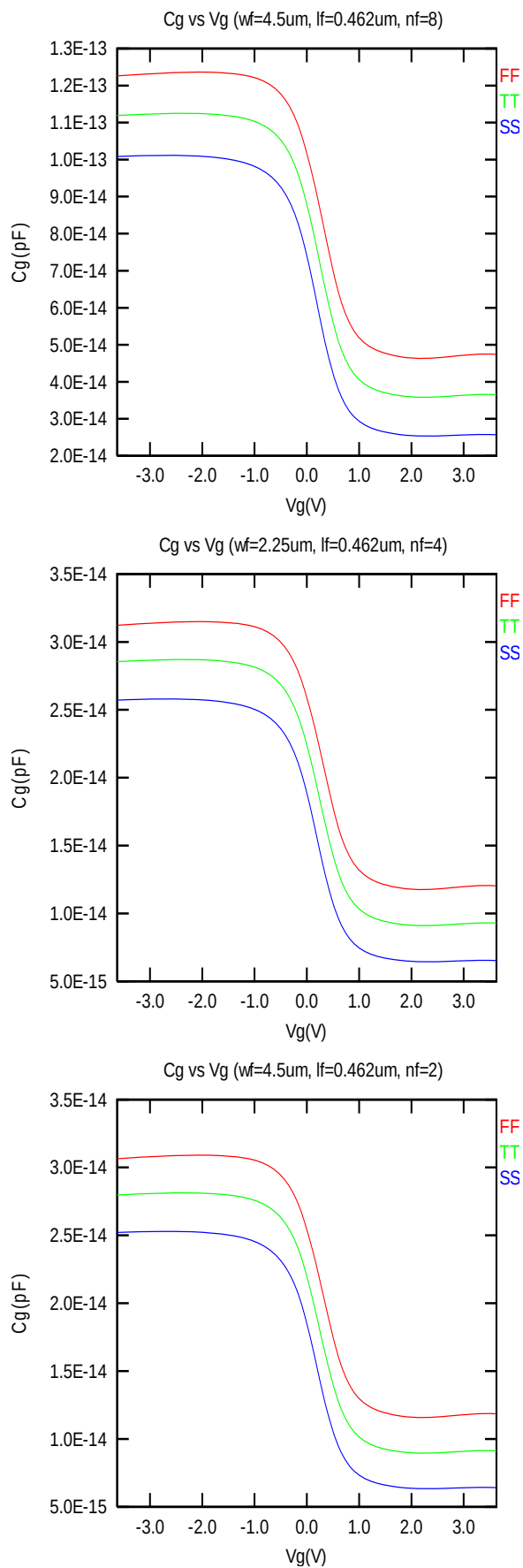


Figure 2-2 Corner C_g vs. V_g

2.3 Equation to Estimate Maximum Capacitance

The following equation can be used to estimate the capacitance value at ($V_{gb}=-3.3V$) at low frequency (100kHz) for given dimension (LF, WF, NF)

$$C=1.0537*5.95e-15*(LF*1e6+0.053)*(WF*1e6+0.042)*NF$$

The unit for capacitance is Farad, for LF & WF is meter.

For example, for a PCAP with LF=1.812um, WF=4.5um, NF=2 (after shrinkage), the capacitance value at $V_{gb}=-3.3V$ is about 1.0621E-13(F).

3 Appendix

3.1 HSPICE Warning Message

Item	Warning Message	Explanation
1.	None.	None.

3.2 SPECTRE Warning Message

Item	Warning Message	Explanation
1.	None.	None.

3.3 ELDO Warning Message

Item	Warning Message	Explanation
1.	None.	None.